

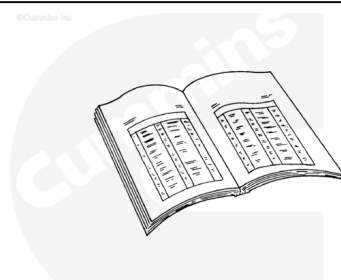
Trainings ▾

Preparatory Steps

Automotive with CM875, CM870 and CM570

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Make sure that electrical power is removed from exhaust gas recirculation (EGR) valve before disconnecting the motor and position sensor connectors from the engine harness. The EGR valve can be damaged.

⚠ CAUTION ⚠

Do not apply 12-VDC directly to the EGR valve motor. Damage will result to the EGR valve.

- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- If necessary, disconnect the air intake piping. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-027-tr.html>)
- Disconnect the EGR valve motor controller connector.
- Disconnect the EGR valve position sensor connector. Refer to Procedure 019-372 in Section 19.

(/qs3/pubsys2/xml/en/procedures/105/105-019-372.html)

- Remove the EGR valve coolant lines. Refer to Procedure 011-030 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-030-tr.html)

Automotive with CM876

⚠ CAUTION ⚠

Make sure that electrical power is removed from exhaust gas recirculation (EGR) valve before disconnecting the motor connector from the engine harness. The EGR valve can be damaged.



Note : Brush away any loose dirt from the vicinity of the air handling connections to avoid contamination of the interior of the engine.

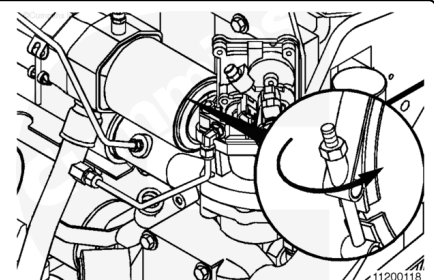
- Disconnect the EGR valve actuator connector.
- Disconnect the EGR temperature sensor. Refer to Procedure 019-378 in Section 19.
(/qs3/pubsys2/xml/en/procedures/101/101-019-378.html)

Remove

Automotive with CM875, CM870 and CM570

Loosen the EGR valve mounting capscrews, but do **not** remove the capscrews.

Loosen the v-band clamp between the EGR cooler and the EGR valve. When removing the v-band clamp, do **not** damage the EGR cooler connection.

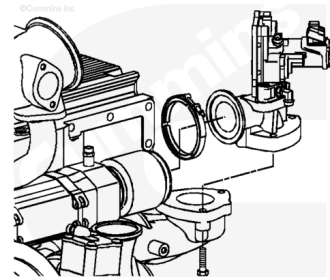


Remove the EGR valve mounting capscrews.

Remove the EGR valve.

Remove the v-band clamp.

Remove and discard the EGR valve gaskets.

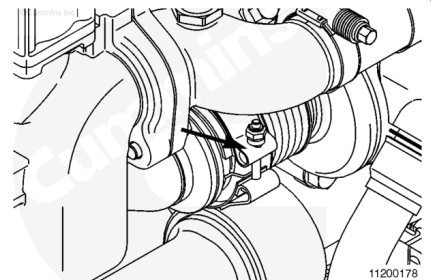


Automotive with CM876

Loosen the v-band clamps at the EGR valve inlet section.

Note : When removing the v-band clamp, do not damage the EGR cooler outlet bellows connection.

The v-band **must** be moved forward and allowed to rest on the bellows protection flange.

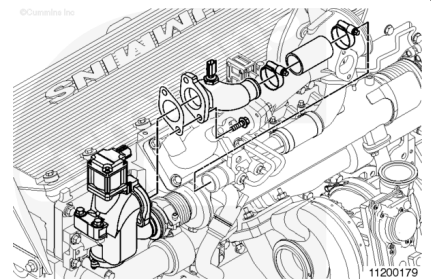


Loosen the EGR valve connection tube mounting capscrews.

Remove the EGR connection outlet tube and hose from the EGR valve and EGR venturi.

Remove and discard the gasket.

After removal of component, cover all open points with protective caps from the Air Handling Clean Care Kit or heavy tape if needed.

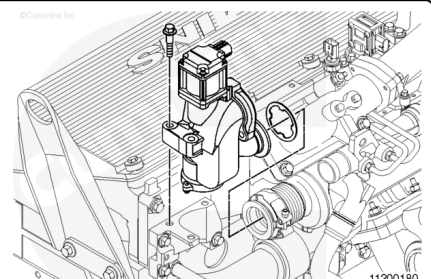


Loosen and remove the three EGR valve mounting capscrews.

Remove the EGR valve and v-band clamps.

After removal of the component, cover all open points with protective caps from the Air Handling Clean Care Kit or with heavy tape, if needed.

Remove and discard the gaskets.



Clean and Inspect for Reuse

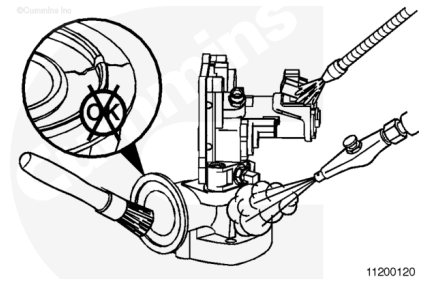
Automotive with CM875, CM870 and CM570



When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Check to be sure the EGR poppet head is present. If the poppet head is missing, run a stiff wire through every port of the exhaust manifold to locate the missing poppet head.

Do **not** assemble the engine without finding the missing poppet head.

Clean the mounting surfaces with a shop towel wetted with safety solvent to remove any deposits or debris.

Dry with compressed air.

Inspect the mounting surfaces for cracks, fretting, or other damage.

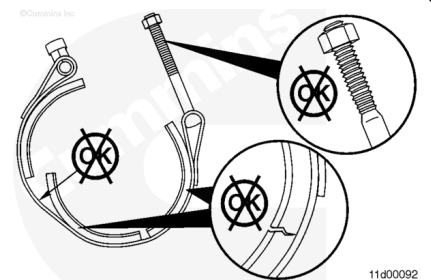
Inspect the EGR valve motor and position sensor for damage.

Replace the EGR valve if damage is found.

Inspect the v-band clamps for signs of over extension. The band **must not** be bent or cracked.

Inspect the v-band clamp threads for damage.

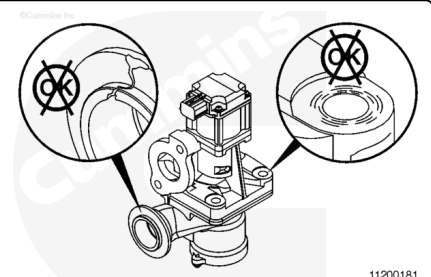
Replace the v-band clamp if damage is found.



Automotive with CM876

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

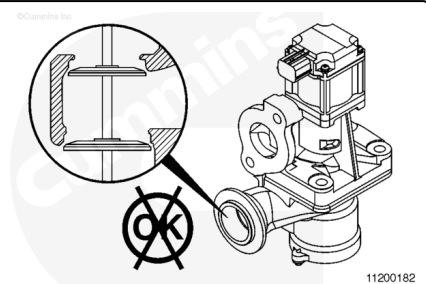
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the EGR valve mounting and sealing surfaces with a shop towel wetted with safety solvent to remove any deposits or debris.

Dry with compressed air.

Inspect the mounting surfaces for cracks, fretting, or any other damage.

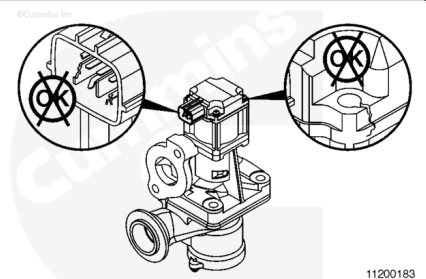
Inspect the EGR valve poppets for a stuck open condition. If the EGR valve is stuck open, replace the EGR valve. Do **not** try to pry the valve poppets open or closed.



Inspect the EGR valve actuator for cracks or any other damage. Replace if damage is found.

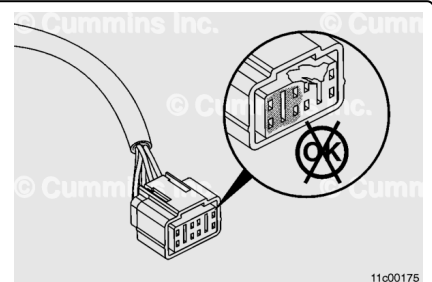
Inspect the EGR valve actuator connector for corrosion, worn pins, or any other damage. Replace the EGR valve assembly if corrosion, worn pins, or any other damage is found.

After inspection of component, cover all open points with protective caps from the Air Handling Clean Care Kit or with heavy tape, if needed.

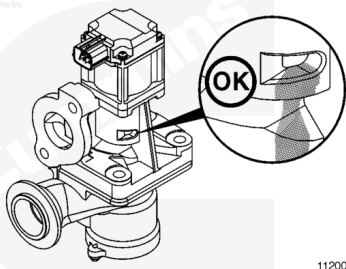


Inspect the engine wiring harness EGR valve actuator connector for corrosion, worn pins, or any other damage.

Replace the engine wiring harness EGR valve connector lead if corrosion, worn pins, or any other damage is found.



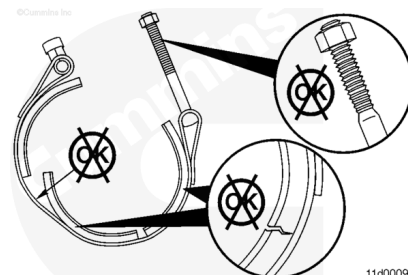
Soot leakage at the EGR valve weep hole is normal and can be expected. Do **not** replace the EGR valve if soot streaks are found at the EGR valve weep hole.



11200185

Inspect the v-band clamps for signs of over extension. The band **must not** be bent or cracked.

Inspect the v-band clamp threads for damage. Replace the v-band clamp if any damage is found.



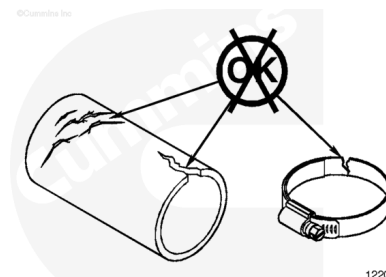
11d00092

Inspect the hose clamp(s) for cracks or other damage.

Replace hose clamp(s) if cracks or other damage is found.

Inspect hose(s) for cracks or other damage.

Replace hose(s) if cracks or other damage is found.



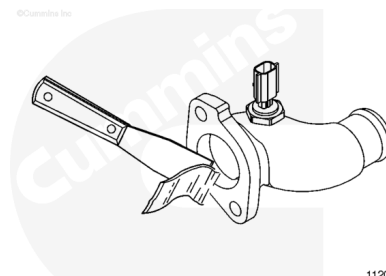
12200023

Clean the remaining gasket material from the EGR connection tube with a gasket scraper.

Clean the surfaces with a shop towel wetted with safety solvent to remove any deposits or debris.

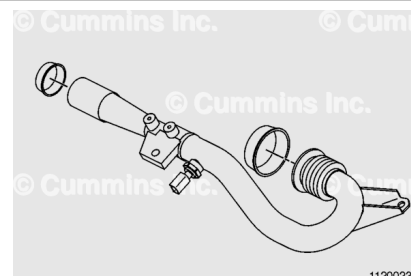
Dry with compressed air.

Inspect the mounting surfaces for cracks, fretting, or any other damage.



11200187

After cleaning and inspection of the component, cover all open points with protective caps from the Air Handling Clean Care Kit or with heavy tape, if needed.

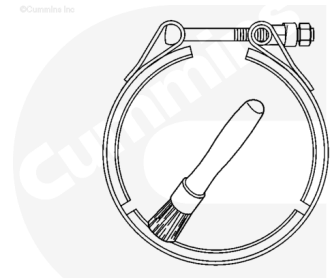


11200236

Install

Automotive with CM875, CM870 and CM570

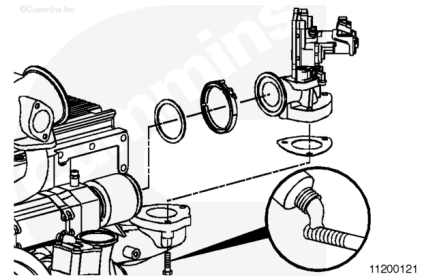
Before installing the v-band clamp, coat the clamping surface with high temperature anti-seize compound.



Install a new EGR valve to EGR cooler gasket.

Install a new EGR valve to exhaust manifold mounting gasket on the exhaust manifold.

Install the v-band clamp around the neck of the EGR valve. Orient the threaded screw to the top of the EGR valve with the nut facing out.



Install the EGR valve and a new gasket onto the exhaust manifold.

Apply a film of high temperature anti-seize compound to the threads of the EGR valve mounting capscrews.

Install the two capscrews installed up through the exhaust manifold and one capscrew installed down through the EGR valve.

Tighten the capscrews.

Torque Value: 54 n•m [40 ft-lb]

Position the v-band clamp over the EGR cooler and EGR valve joint as illustrated in the graphic.

Tighten the v-band clamp.

Torque Value: 17 n•m [150 in-lb]

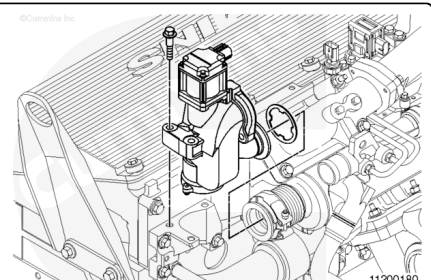
Automotive with CM876

Remove the protective caps or heavy tape.

Install a new EGR valve to EGR cooler gasket.

Install the EGR valve onto the EGR valve mounting bracket.

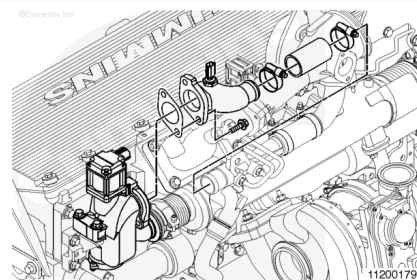
Install a new EGR valve connection tube gasket onto the EGR connection tube.



Lubricate the EGR connection hose end and the EGR venturi and loosely install two hose clamps.

Coat the clamping surface with high temperature anti-seize compound, before installing the v-band clamp.

Rest the v-band clamp on the EGR cooler bellows protecting shield, with the nut positioned properly for adequate access in chassis.



Install the EGR connection tube onto the EGR valve and properly position the hose over the EGR connection and EGR valve outlet.

Install the three capscrews (1) installed into the EGR valve mounting bracket.

Tighten the mounting capscrews (1).

Torque Value: 54 n•m [40 ft-lb]

Install the two EGR connection mounting capscrews (2) into the EGR valve.

Torque Value: 54 n•m [40 ft-lb]

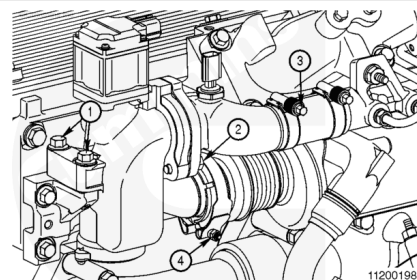
Properly torque the EGR connection tube hose clamps (3).

Refer to Procedure 011-025 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)

Tighten the EGR valve inlet v-band clamp (4).

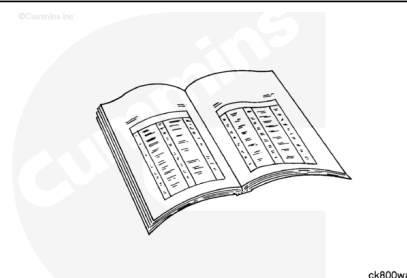
Torque Value: 17 n•m [150 in-lb]



Finishing Steps

Automotive with CM875, CM870 and CM570

- Install the EGR valve coolant lines. Refer to Procedure 011-030 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-030-tr.html)
- Connect the EGR valve position sensor connector. Refer to Procedure 019-372 in Section 19.
(/qs3/pubsys2/xml/en/procedures/105/105-019-372.html)



- Connect the EGR valve motor connector.
- Install the air intake piping if necessary. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-027-tr.html>)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Start and operate the engine and check for exhaust or coolant leaks.

Automotive with CM876

- Connect the EGR temperature sensor. Refer to Procedure 019-378 in Section 19. (</qs3/pubsys2/xml/en/procedures/101/101-019-378.html>)
- Connect the EGR valve actuator connector.
- Start and operate the engine and check for exhaust or coolant leaks.
- If the damaged component resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14. (</qs3/pubsys2/xml/en/procedures/101/101-014-013.html>)

